

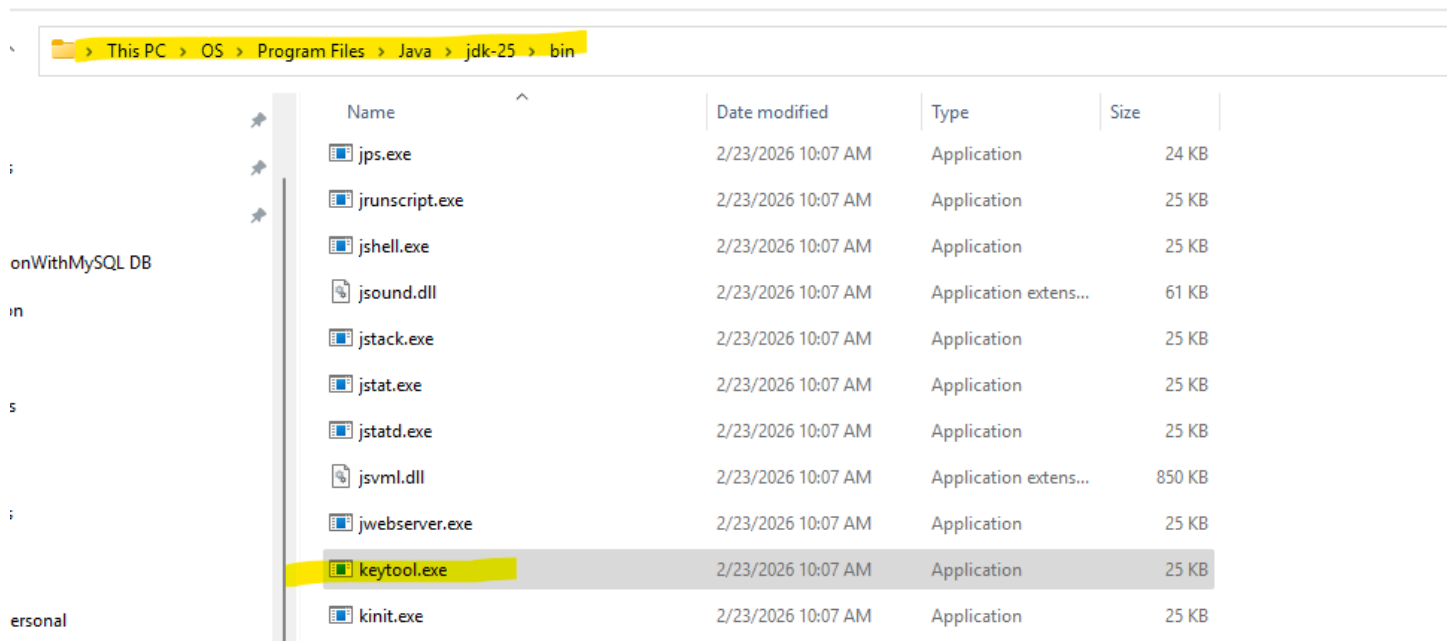
Using Keystore

- Is the JDK creation

What Is Keytool and Why Use It?

→ *Keytool is a command-line utility provided with the Java Development Kit (JDK)*

→ *It helps you create and manage keystores, generate key pairs (RSA keys, for example), and handle digital certificates.*



Name	Date modified	Type	Size
jps.exe	2/23/2026 10:07 AM	Application	24 KB
jrunscript.exe	2/23/2026 10:07 AM	Application	25 KB
jshell.exe	2/23/2026 10:07 AM	Application	25 KB
jsound.dll	2/23/2026 10:07 AM	Application extens...	61 KB
jstack.exe	2/23/2026 10:07 AM	Application	25 KB
jstat.exe	2/23/2026 10:07 AM	Application	25 KB
jstatd.exe	2/23/2026 10:07 AM	Application	25 KB
jsvml.dll	2/23/2026 10:07 AM	Application extens...	850 KB
jwebserver.exe	2/23/2026 10:07 AM	Application	25 KB
keytool.exe	2/23/2026 10:07 AM	Application	25 KB
kinit.exe	2/23/2026 10:07 AM	Application	25 KB

- we use keystore ដោយសារតែយើងមិនចាំបាច់ សរសេរកូដ ខ្លួនឯង

How to generate code command:

>>

Explain command:

keytool -genkeypair

-alias config-server-key ដាក់ឈ្មោះអោយ key នឹងឈ្មោះអី

-keyalg RSA ប្រើប្រាស់ Algorithm អី

-dname "CN=Config Server,OU=Spring Cloud,O=Company" (destingvice) ការកំណត់ឈ្មោះអោយវា

-keypass mypass លេខសំខាត់សម្រាប់ key នឹង(config-server-key)

-keystore config-server.jks រក្សា លេខសំខាត់ (កន្លែងដែលវាបង្កើតហើយ ដាក់វានៅកន្លែងណា location) ex: D://config-server.jks

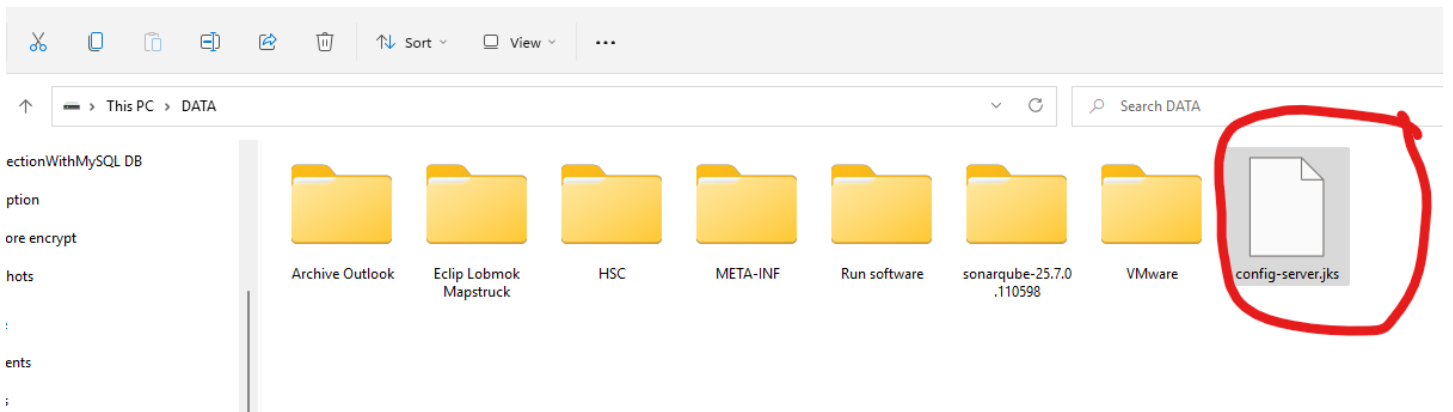
-storepass mypass លេខសំខាត់សម្រាប់ open key (config-server.jks) ប្រើសម្រាប់ open keystore

```
C:\Windows\System32\cmd.exe

C:\Program Files\Java\jdk-25\bin>keytool.exe -genkeypair -alias config-server-key -keyalg RSA -dname "CN=Config Se
rver,OU=Spring Cloud,O=Company" -keypass mypass -keystore D:\config-server.jks -storepass mypass
Generating 3072-bit RSA key pair and self-signed certificate (SHA384withRSA) with a validity of 90 days
for: CN=Config Server, OU=Spring Cloud, O=Company

C:\Program Files\Java\jdk-25\bin>
```

>> keytool.exe -genkeypair -alias config-server-key -keyalg RSA -dname "CN=Config Server,OU=Spring Cloud,O=Company" -keypass mypass -keystore D:\config-server.jks -storepass mypass



It will save to the D drive: then we can copy to the source in java or we directly to that location.

Application.yaml:

encrypt:

key-store:

location: config-server.jks

password: mypass

alias: config-server-key

```
33 encrypt:
34 key-store:
35   location: config-server.jks
36   password: mypass
37   alias: config-server-key
```

- Encrypt

The screenshot shows a REST client interface for a POST request to `http://localhost:8888/encrypt`. The request body is raw and contains a long, complex string of characters. The response status is 200 OK, with a response time of 77 ms and a body size of 765 B. The response body is also raw and contains a long, complex string of characters. A yellow highlight is drawn around the response body.

```
POST http://localhost:8888/encrypt

Database-Development-Build SERVER DB

AYC6SIKURiMexh96Gzb+MpNq1o4g6UsBXUQooA+K2aR+4Sg0B08TwzCoGbBdCCLCYdYBtg6WURus10HI5UyTD4NTszFshcSvQe/KTtFHE1/JQip3qFk+671UQU1XJ37CBQsIzVMKNoIKTrUPXVuhHf1
+1Te1x2uJB3o/HEXdTW+iFhRkzHn4gsaJoo3VcNgDqEdgaHr+cv3aNs2o9PvPsBKJgJ2IywwJ0b1cckJJUSShq07jGAtt0Uq630MaHSbpy/Pc8jzxbY
+LIW9mgRwMpJqbuozQW2648eVnzjfwWkSHJALVgzdDRT5Am5j8UUh4y1Xy0JrHY080x80h3Vdz0/bXGhonJ2rgc8AwkQ2+qSfUgX3woSSw+1jIwYWe/Yz0
+30o9Gtu3mkM6sNY3HaMW9mV8RwhpX5aCwABDepmX50Pak9e2sFchG00dzNqqWDWjZhXBeEDe6MQNEBS8T6SsYwa4+kDYU8TPeF0ZgM+5dsA3qS/hLo0LtlfiJEB
+3xB03iGcoB1uI3pzi0GMzwbZ6gRxx+/hmcV+X590qfGVvPU10qyFxxCnda1P7jS9eIncBuzPXzqw0sy80y0/qze3V6tA
```

- Decrypt

The screenshot shows a REST client interface for a POST request to `http://localhost:8888/decrypt`. The request body is raw and contains a long, complex string of characters. The response status is 200 OK, with a response time of 46 ms and a body size of 200 B. The response body is raw and contains the text `Database-Development-Build SERVER DB`, which is highlighted with a yellow circle.

```
POST http://localhost:8888/decrypt

AYC6SIKURiMexh96Gzb+MpNq1o4g6UsBXUQooA+K2aR+4Sg0B08TwzCoGbBdCCLCYdYBtg6WURus10HI5UyTD4NTszFshcSvQe/KTtFHE1/JQip3qFk+671UQU1XJ37CBQsIzVMKNoIKTrUPXVuhHf1
+1Te1x2uJB3o/HEXdTW+iFhRkzHn4gsaJoo3VcNgDqEdgaHr+cv3aNs2o9PvPsBKJgJ2IywwJ0b1cckJJUSShq07jGAtt0Uq630MaHSbpy/Pc8jzxbY
+LIW9mgRwMpJqbuozQW2648eVnzjfwWkSHJALVgzdDRT5Am5j8UUh4y1Xy0JrHY080x80h3Vdz0/bXGhonJ2rgc8AwkQ2+qSfUgX3woSSw+1jIwYWe/Yz0
+30o9Gtu3mkM6sNY3HaMW9mV8RwhpX5aCwABDepmX50Pak9e2sFchG00dzNqqWDWjZhXBeEDe6MQNEBS8T6SsYwa4+kDYU8TPeF0ZgM+5dsA3qS/hLo0LtlfiJEB
+3xB03iGcoB1uI3pzi0GMzwbZ6gRxx+/hmcV+X590qfGVvPU10qyFxxCnda1P7jS9eIncBuzPXzqw0sy80y0/qze3V6tA

Database-Development-Build SERVER DB
```

- Configure to the GitHub

- Application.yaml

```
spring:
  application:
    name: configserver
  cloud:
    config:
      server:
        git:
          uri: https://github.com/longvibol/app-configuration.git
          username: longvibol
          password: ${GIT_TOKEN}
          default-label: main
          clone-on-start: true
```

```
encrypt:
  key-store:
    location: config-server.jks
    password: mypass
    alias: config-server-key
```

```
application.yaml X
1 server:
2   port: 8888
3
4 spring:
5   application:
6     name: configserver
7   cloud:
8     config:
9       server:
10      git:
11        uri: https://github.com/longvibol/app-configuration.git
12        username: longvibol
13        password: ${GIT_TOKEN}
14        default-label: main
15        clone-on-start: true
16
17 encrypt:
18   key-store:
19     location: config-server.jks
20     password: mypass
21     alias: config-server-key
```

app-configuration / configdemo-prod.yml in main

Cancel changes

Commit changes...

Edit Preview

Spaces

2

No wrap

```
1 build:
2   id: 101
3   version: 1.2.3
4   name: "{cipher}JAYDGDtzRVZ5f/CmFMH9lmPxs3PqkMDA3/1EYxDOamvvNIe1DKqIFkphRmHfNEHexrqNKjrS0SuIuTtj06XQgVnkq/hFn4wfBoHqah1dfw48nAoxtd07UyE+dXgaL+imItT6e1hrSTsXRT7j/68Ldt01ie7CcnzEonNxueV8StECrtX0QewYS"
5   type: "GIDHUB: Production Build"
6
```